Ontology Learning

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Outline

1. Introduction
2. Methods
   - Ontology Learning from Text
     - Terms
     - Synonyms
     - Concepts
     - Taxonomy
     - Relations
     - Rules and Axioms
   - Ontology Learning from Folksonomies
3. Tools
4. Conclusion

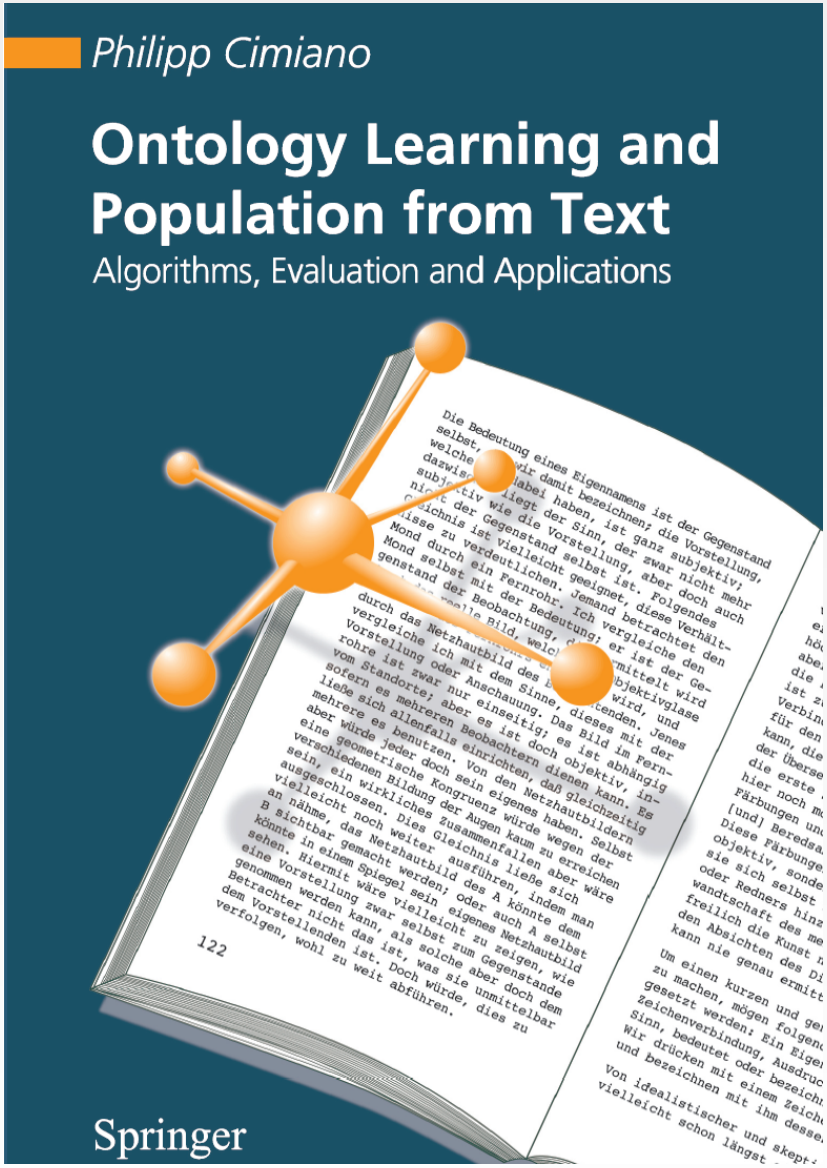
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Too many names, the same subject

- Ontology
  - Extraction
  - Emergence
  - Generation
  - Acquisition
  - Discovery
  - Population
  - Enrichment

Ontology Learning!



(Cimiano, 2006)

WHAT is Ontology Learning (OL)?

- Methods and techniques for (OntoSum, 2008):
  - Building an ontology from scratch
  - Enriching, or adapting an existing ontology
- Extract concepts and relations to form an ontology (Wikipedia, 2008a)
- OL is a semi-automatic task of information extraction

What is Ontology Learning for? (WHY)

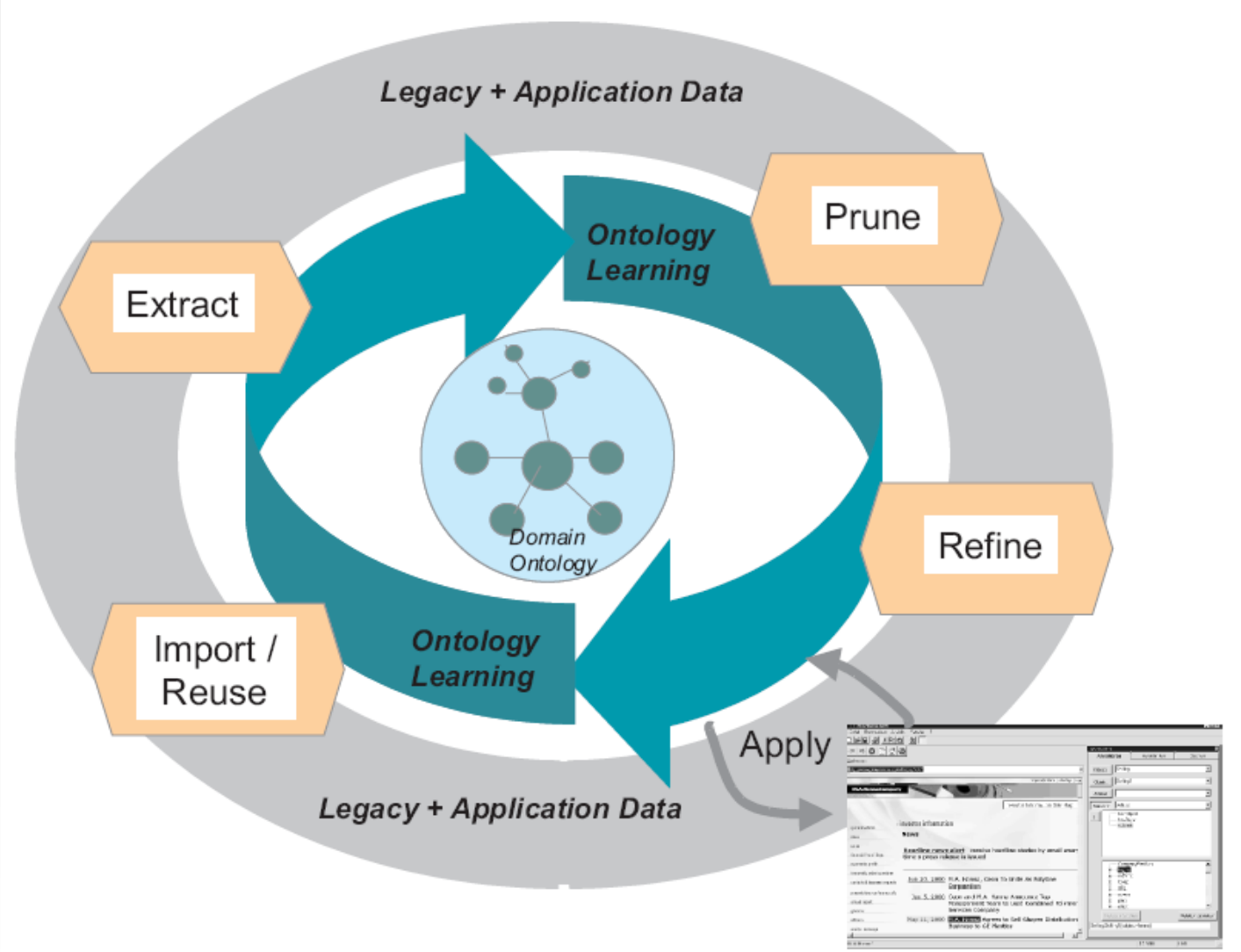
- Problems in Ontology Engineering (OE) (Maedche and Staab, 2001):
  - Can you develop an ontology fast? (time)
  - Is it difficult to build an ontology? (difficulty)
  - How do you know that you've got the ontology right? (confidence)
- OL can overcome these problems, specially the Knowledge Acquisition bottleneck

Information Sources

- Relevant text (Web documents mainly)
- Web document schemata (XML, DTD, RDF)
- Databases on the Web
- Dictionaries
- Semi-structured documents
- Personal Wikis, e-mail/file folders
- Existing Web ontologies

OE Cycle (Maedche and Staab, 2001)

- OL is not only the task of extraction



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How to Learn Ontologies?

- Natural Language Processing
- Dictionary Parsing
- Statistical Analysis
- Machine Learning
- Hierarchical Concept Clustering
- Formal Concept Analysis (Lattices)

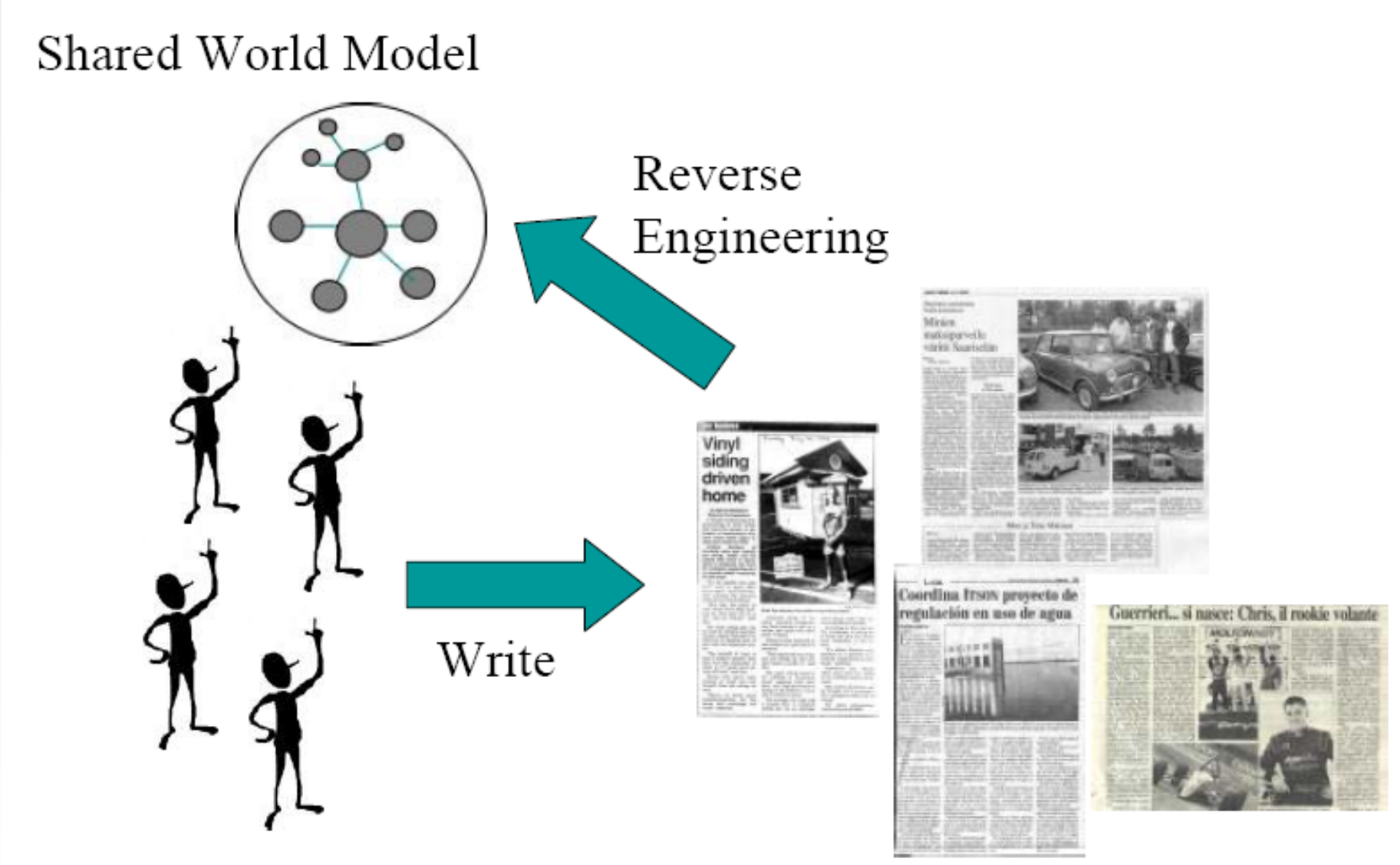
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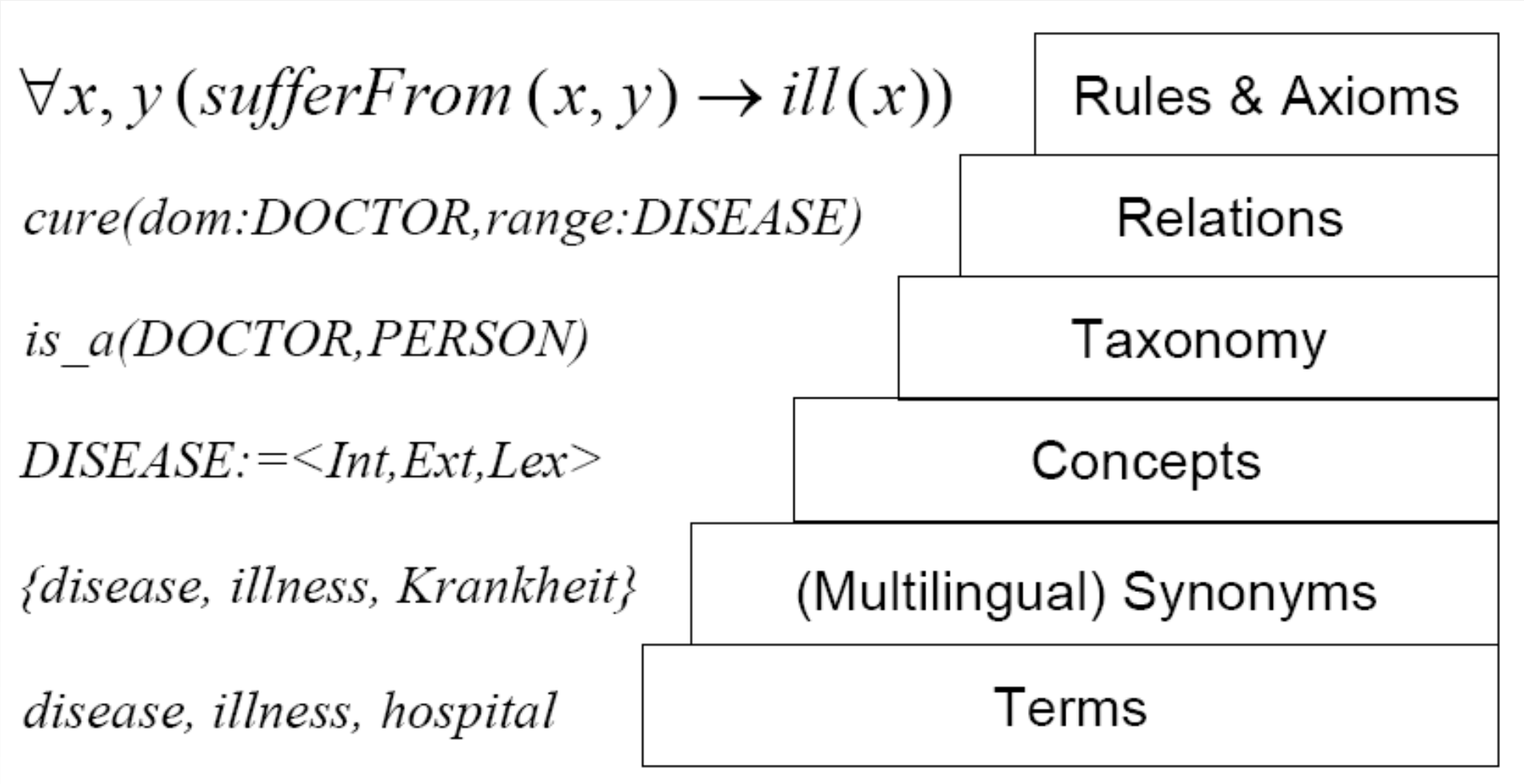
Why Text?

- Text is massively available on the Web
- Relevant texts contain relevant knowledge about a domain
- Linguistic knowledge remains associated with the ontology (Sintek et al., 2004)

OL as Reverse Engineering (Buitelaar et al., 2005)



OL from Text Layer Cake (Buitelaar et al., 2005)



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Term Extraction - Linguistic Methods

- Part-of-speech tagging: Identify syntactic class
  - Ex: Noun -> Class, Verb -> Relation
- Stemming
  - Ex: **Formal**(ize/ization/ized/izing)
- Head-modifier analysis
  - Ex: Fast **car**, the hood of the **car**
- Grammatical function analysis
  - Ex: “John played football in the garden” -> play(John,football)

Term Extraction - Other methods

- Statistical Methods
  - Term Weighting (TF-IDF)
  - Co-occurrence analysis (Common method applied in Text Mining)
  - Comparison of frequencies between domain and general corpora
- Hybrid Methods
  - Linguistic rules to extract term candidates
  - Statistical (pre- or post-) filtering

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Synonym Extraction

- Extending WordNet (Term Classification)
- Co-occurrence between terms (Term Clustering)

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Concept Extraction

A term may indicate a concept, if we define its:

- Intension
  - (In)formal definition of the objects this concept describes
  - Ex: A **disease** is an impairment of health or a condition of abnormal functioning
- Extension
  - Set of objects described by this concept
  - Ex: Cancer, heart disease
- Lexical Realizations
  - The term itself and its multilingual synonyms

Intension

- Informal definition - a shallow definition as used in WordNet
  - Find the appropriate WordNet concept for a term and the appropiate conceptual relations (Navigli and Velardi, 2004)
- Formal definition - formal constraints defining class membership
  - Formal Concept Analysis

Extension

- Extraction of instances for a concept from text (Ontology Population)
- Relates to Knowledge Markup and Tag Suggestion (Semantic Metadata)
- Use Named-Entity Recognition
  - Ex: John is a football player -> **John** (Person) is an instance of **Football Player**
- Instances can be:
  - Names for objects
    - Ex: Person, Organization, Country, City
  - Event instances
    - Ex: Football Match (with Teams, Players, Officials, etc)

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